Computers in Biology and Medicine 164 (2023) 107251

Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed

Improving adversarial robustness of medical imaging systems via adding global attention noise

Yinyao Dai [a], Yaguan Qian [a,*], Fang Lu [a], Bin Wang [b,*], Zhaoquan Gu [c], Wei Wang [d], Jian Wan [a], Yanchun Zhang [e]

[a] Zhejiang University of Science and Technology, Hangzhou 310023, China
[b] Zhejiang Key Laboratory of Multidimensional Perception Technology, Application, and Cybersecurity, Hangzhou 310052, China
[c] School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518071, China
[d] Beijing Key Laboratory of Security and Privacy in Intelligent Transportation, Beijing Jiaotong University, Beijing 100091, China
[e] Victoria University, Melbourne, VIC 8001, Australia

ARTICLE INFO

Keywords:
Medical image
Model robustness
Noise injection
Adversarial attack

ABSTRACT

Recent studies have found that medical images are vulnerable to adversarial attacks. However, it is difficult to protect medical imaging systems from adversarial examples in that the lesion features of medical images are more complex with high resolution. Therefore, a simple and effective method is needed to address these issues to improve medical imaging systems' robustness. We find that the attackers generate adversarial perturbations corresponding to the lesion characteristics of different medical image datasets, which can shift the model's attention to other places. In this paper, we propose global attention noise (GATN) injection, including global noise in the example layer and attention noise in the feature layers. Global noise enhances the lesion features of the medical images, thus keeping the examples away from the sharp areas where the model is vulnerable. The attention noise further locally smooths the model from small perturbations. According to the characteristic of medical image datasets, we introduce Global attention lesion-unrelated noise (GATN-UR) for datasets with unclear lesion boundaries and Global attention lesion-related noise (GATN-R) for datasets with clear lesion boundaries. Extensive experiments on ChestX-ray, Dermatology, and Fundoscopy datasets show that GATN improves the robustness of medical diagnosis models against a variety of powerful attacks and significantly outperforms the existing adversarial defense methods. To be specific, the robust accuracy is 86.66% on ChestX-ray, 72.49% on Dermatology, and 90.17% on Fundoscopy under PGD attack. Under the AA attack, it achieves robust accuracy of 87.70% on ChestX-ray, 66.85% on Dermatology, and 87.83% on Fundoscopy.

1. Introduction

Recent advances in deep neural networks (DNNs) have achieved remarkable success in medical diagnosis tasks, such as lung diseases classification from ChestX-ray [1], skin cancer classification from Dermatology [2], and diabetic retinopathy classification from Fundoscopy [3]. However, a large body of work on natural images has revealed that the classification accuracy of DNNs significantly degrades under adversarial attacks. The input examples are imposed with elaborate but human-imperceptible perturbations to fool DNNs into making incorrect predictions with high confidence [4,5]. Unfortunately, medical images are more vulnerable to adversarial attacks than natural images [6–8]. In medical imaging systems, such a vulnerability will cause large economic loss and serious security risks [9]. Therefore, improving the robustness of medical diagnosis models under adversarial attacks is critical.

In the field of natural images, Kalimeris et al. [10] show that with the continuous training of the network, a significant increase in the curvature of the decision boundary and loss landscape will occur, and the adversarial examples are easy to hide in these isolated regions with high curvature [11]. Then, Ilyas et al. [12] raise a new explanation for the existence of adversarial examples. The vulnerability of the model arises from being more sensitive to features that generalize better in the data, *i.e.*, small changes to these features can dramatically alter the outputs of the model. This means that the intrinsic features of the example are stable and the generalization features are unstable. The attacker easily causes the model to misclassify by adding perturbations to the generalization features, thus pushing the examples into the sharp regions of the model.

Numerous methods have been proposed to defend against adversarial attacks in the field of natural images [13–16] and medical

* Corresponding authors.
E-mail addresses: qianyaguan@zust.edu.cn (Y. Qian), wbin2006@gmail.com (B. Wang).

https://doi.org/10.1016/j.compbiomed.2023.107251
Received 17 April 2023; Received in revised form 14 June 2023; Accepted 7 July 2023
Available online 11 July 2023
0010-4825/© 2023 Elsevier Ltd. All rights reserved.

images [17–20]. However, these methods have some limitations. First, most of them are defenses against specific attacks, which means that they cannot protect the model from unknown attacks in the future. Second, these methods sacrifice clean accuracy. Third, the defense methods in the medical image domain do not adequately evaluate advanced attack methods, such as AutoAttack (AA). Finally, the existing defense methods ignore the intrinsic properties of different medical image datasets.

In this paper, we propose a simple but effective defense method to improve model robustness by global attention noise (GATN) injection. In particular, we make the following observation that medical image adversarial examples can also shift the model's attention to unstable generalization features. In addition to this, adversarial examples from different medical image datasets distract the model's attention to different degrees. To be specific, for the Dermatology and Fundoscopy datasets the model's attention to the adversarial examples remains around the lesion areas. In contrast, for the ChestX-ray dataset with clearly exploitable generalization features, the model's attention is completely distracted. Then, to effectively defend against adversarial attacks, we address this problem from both the example and model's perspectives. First, we add global noise to the example layer. Since the example layer contains all information, adding trainable random noise to the example layer can strengthen the stability of the essential features of the examples and weaken the instability of the generalization features, thus keeping the examples away from the sharp regions of the model. Second, we add attention noise to the feature layers. This noise is added to the lesion-related or lesion-unrelated regions during training, according to the characteristics of the medical image datasets. Attention noise can smooth the model while highlighting salient areas. Despite its simplicity, our method can improve robust accuracy with the smallest possible sacrifice of generalization accuracy.

Our main contribution in this paper is summarized as follows:

- We propose global attention noise (GATN) injection and introduce two feature layer noises (GATN-R and GATN-UR) to enhance the noise representation. We add trainable global noise in the example layer to strengthen the essential features of the examples and weaken the instability of the generalization features, thus pushing the examples away from the sharp regions. Trainable attention noise is added in the feature layers to locally smooth the model and highlight the salient regions. Both can strengthen the resistance of the model to small perturbations.
- The method requires only clean examples, which can reduce the computational cost of generating medical image adversarial examples. Meanwhile, GATN can be trained like other DNNs' parameters to achieve the best performances for the noise. Therefore, robust accuracy can be improved with minimal sacrifice of clean accuracy.

The rest of this paper is organized as follows. In Section 2, we present existing adversarial attacks and defense methods in the field of natural and medical images. In Section 3, we analyze medical image adversarial examples, and global attention noise injection is proposed to defend against adversarial examples. Section 4 presents the benchmark evaluation results of the above methods under various attack environments. We explore the effect of GATN on the model in terms of loss landscape and human vision alignment in Section 5. Finally, we discuss the current limitations and future research directions in Section 6.

2. Related work

2.1. Adversarial attack

Despite the excellent performance of DNNs in medical imaging systems, Szegedy et al. [4] first discover the limitations of DNNs to adversarial examples. An adversarial example is a carefully crafted image to fool the target model by adding small adversarial perturbations to the original clean image, and due to its transferability, the perturbation designed for one network can also be used to fool other networks. Recently, adversarial attacks in the natural domain have been developed rapidly. Goodfellow et al. [21] propose an efficient single-step attacking method named Fast Gradient Sign Method (FGSM) that perturbs the pixels of inputs according to the direction of the gradient under a L_p perturbation budget. Kurakin et al. [22] propose Basic Iterative Method (BIM), an iterative version of FGSM, iteratively perturbs the inputs with a smaller step size. Madry et al. [15] update BIM by randomly initializing the input point, which is one of the strongest first-order attacks named Projected Gradient Descent (PGD). Furtherly, Croce et al. [23] integrated multiple attack methods and proposed AutoAttack (AA). Apart from attacks based on gradient, Carlini and Wagner (C&W) [24] formally poses the generation of adversarial example as an optimization problem. Deepfool [25], on the other hand, finds the minimum perturbation through the decision boundary of the model.

There are also some researches on medical adversarial attack. Qi et al. [26] propose Stabilized Medical Image Attacks (SMIA). The loss function can be designed to apply to different types of medical images. Yao et al. [27] propose Hierarchical Feature Constraint (HFC) attack which can bypass detection by hiding features in a normal distribution. Zhou et al. [28] develop two different resolution GAN-based models for breast cancer assisted diagnosis. GAN was used to generate or remove lesion features to generate high-confidence adversarial examples that further induce incorrect diagnosis of breast cancer. Wang et al. [29] propose an attention attack with feature space constraints to efficiently generate adversarial examples. An attention mechanism constraint was proposed to ensure that the adversarial examples are close to the classification boundary of the feature space. Other studies [7,8,30] evaluated the robustness of deep diagnosis models on different tasks by adversarial attacks.

2.2. Adversarial defense

A large number of defense methods are emerging, which can be divided into example-level defense and model-level defense. Example-level defense includes destroying perturbation structures [14,18,31–33] and removing perturbations [34–38]. Xie et al. [14] use random resizing and padding to process input images and feed them into a model for training. Jia et al. [35] propose Comdefend, which includes a compression network and a reconstruction network to extract key features from the adversarial example. Kansal et al. [37] extended the High-level representation Guided Denoiser (HGD) [34], where the guidance of high-level information can further facilitate the elimination of adversarial influences on the final diagnosis. Xu et al. [38] introduce a learnable adversarial denoising method using U-Net [39] as a defender model. The defender model preprocesses the input medical images before feeding them into medical segmentation models. Xu et al. proposed The Medical Retraining Diagnostic Framework (MedRDF) [18]. MedRDF creates various copies of the input image that are each scrambled by isotropic noise. These copies are predicted by majority voting after denoising by a custom denoiser. Wang et al. [33] propose reverse engineering of the adversarial perturbations in skin cancer images. Skin images of different sizes are gradually diffused by injecting isotropic Gaussian noise to move the adversarial examples to clean image manifolds. However, the lesion area of medical images occupies only a few pixels and the location of the lesion area is not fixed [40], using above methods may lead to the loss of lesion features. This affects the classification and defense effect.

Model-level defense is mainly robust training, including adversarial training [15,41–45] and random smoothing [17,46–48]. Adversarial training can be viewed as data augmentation, where adversarial examples are used as training sets to improve model robustness. Han et al. [44] introduce dual-batch normalization to adversarial training,

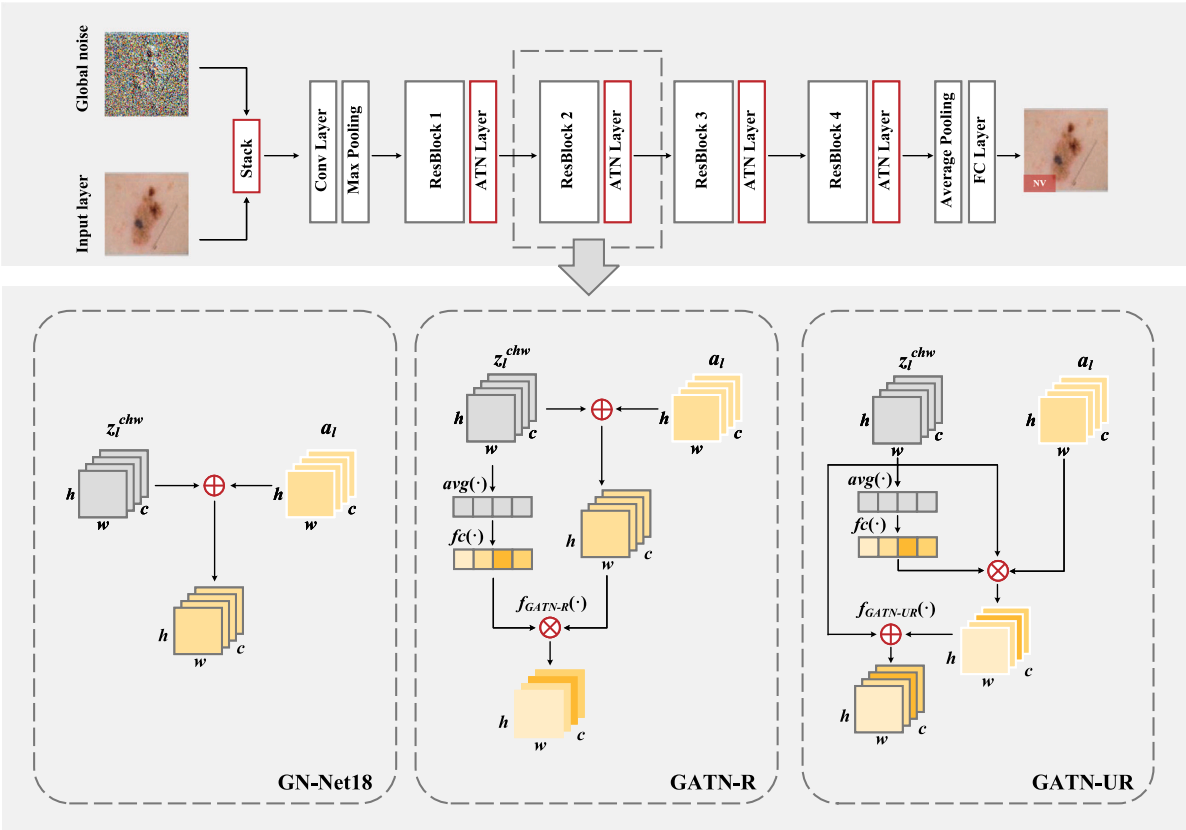


Fig. 1. Framework of our GATN method, which has a backbone network, global noise of example layer, and three types of feature layers noise. GN-Net18 represents global noise. GATN-R and GATN-UR are our proposed lesion-related noise and lesion-unrelated attention noise.

which greatly improved the robustness of the diagnostic model without reducing clean accuracy. Shi et al. [49] embed sparse denoising operators in DNNs to improve robustness by removing noisy features unconsciously learned by DNNs. Wang et al. [45] propose the Multiple Instance Robust Self-Training Method with Drop-Max layer (MIRST-DM) to learn a smooth decision boundary for robust classification of breast cancer. However, adversarial training is not friendly on large datasets and requires a large computational cost to generate adversarial examples. Random smoothing, on the other hand, smoothes the loss landscape of the model by adding small perturbations to reduce the sharp regions. Xue et al. [17] propose an adversarial training approach for medical images by adding fixed Gaussian noise to the input layer and embedding an auto-encoder to keep the deep features constant to stabilize the neural network. Additionally, some research have discovered that users' privacy is easily in danger [50,51]. EMRs (Electronic Medical Records) have taken the place of paper-based medical records. As a result, privacy protection in medical field is likewise becoming a topic of increased interest [52,53].

Compared with the above methods, our approach has two components, combining both example and model levels. From the example level, global noise is added to the example layer to destroy the perturbation structure while preserving the lesion features of medical images. From the model level, we add attention noise to feature layers which locally smooth the model and avoiding the degradation of classification accuracy brought by oversmoothing.

3. Methods

We will first describe the motivation and our method at a high level here. According to previous research, the complex texture of medical images makes them more vulnerable and difficult to train. In Section 3.1, we gain deep insight into medical adversarial images and uncover some intriguing phenomena. Based on above observations, we propose global attention noise injection based on the characteristics of different medical image datasets, as shown in Fig. 1. In our approach, we first add global noise to the example layer to strengthen the stable essential features and weaken the instability of generalization features, thus pushing the examples away from the sharp region of the model (Section 3.2). We add attention noise to the feature layers, adding noise to lesion-related or lesion-unrelated regions according to the characteristics of the medical image datasets, thus locally smoothing the model and highlighting salient features (Section 3.3).

3.1. Discovery

Some studies [34,54,55] have shown that adversarial perturbations can lead to noise in the feature layers. These feature perturbations are amplified gradually along with forward propagation and eventually lead to the wrong prediction. We further explore the adversarial examples in medical images and obtain two discoveries as follows.

First, medical images adversarial examples divert the attention of the model, which eventually leads to classification errors. We use Grad-CAM [56] to visualize the regions of the input examples that have the most significant impact on the output. As shown in Fig. 2, the model's attention is severely diverted. The adversarial examples from the ChestX-ray dataset shift the model's attention from the lesion area to the abdomen and head. The Dermatology dataset adversarial examples shift the model's attention from one side of the lesion region to the other. The adversarial examples from the Fundoscopy dataset, on the other hand, use perturbations to simulate lesions, allowing the model to increase its attention to the pseudo-lesion region. The essential features in medical images are lesion features, and thus complex texture features become generalized features that can be easily exploited by attackers. In addition, we output the average activation values of clean and

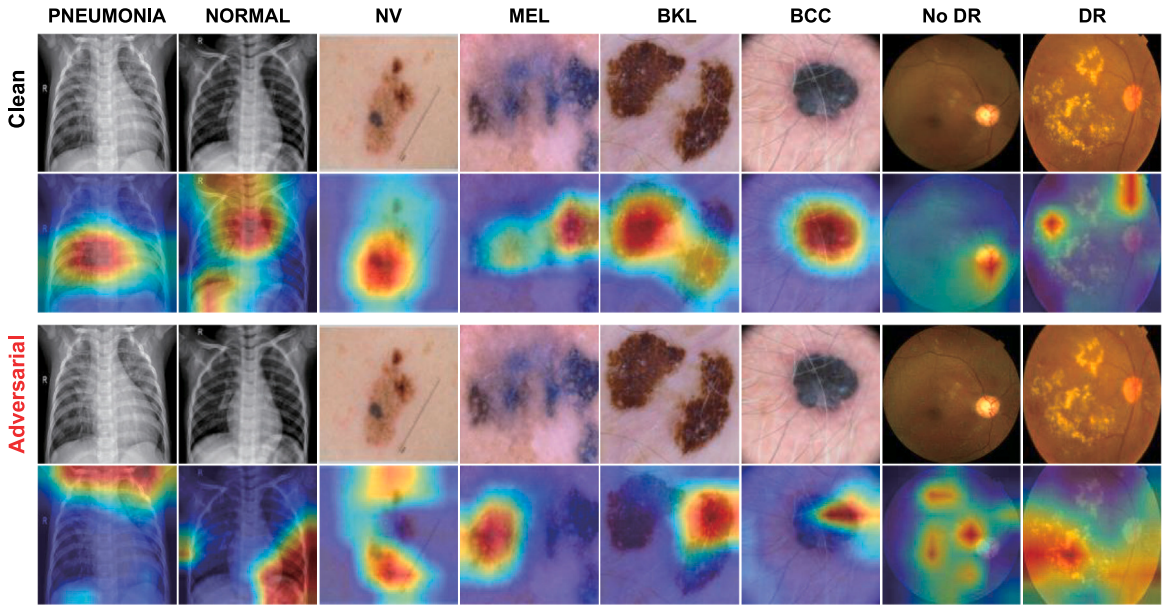


Fig. 2. The attention maps of ResNet18 on normal images (first row) versus adversarial images (third row). The attention maps are computed by the Grad-CAM technique.

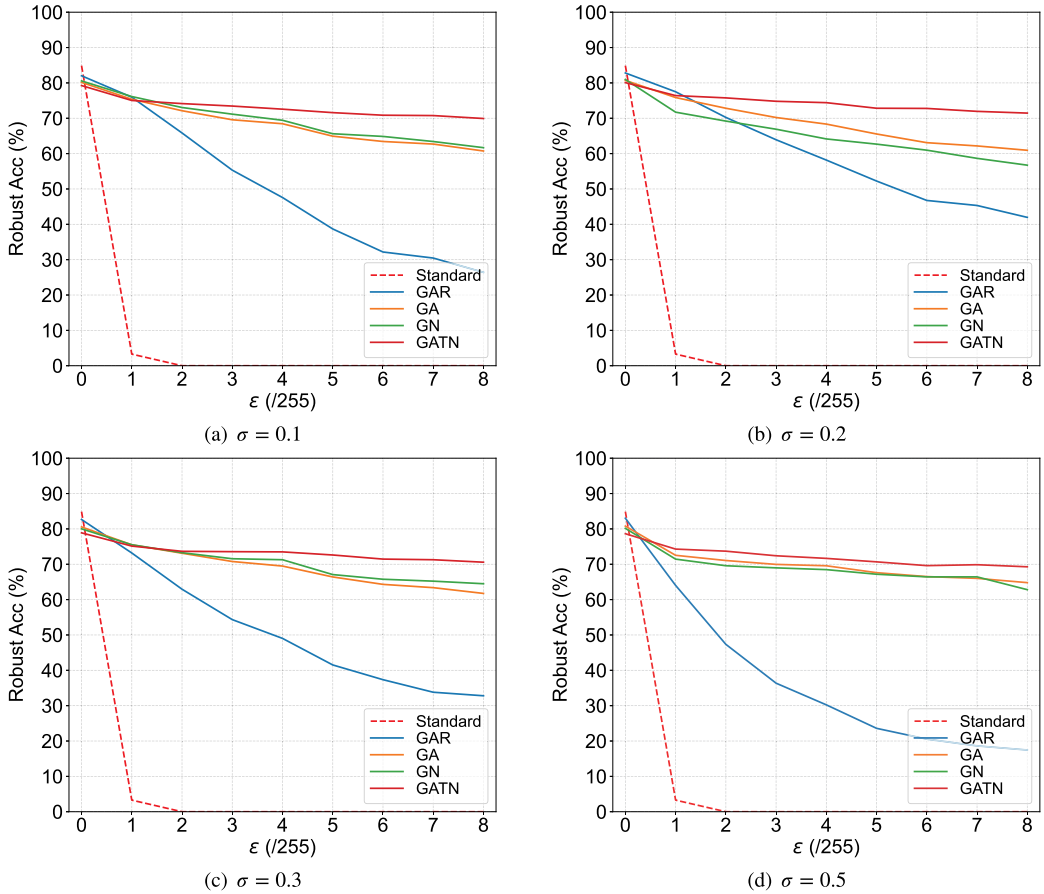


Fig. 3. The robust accuracy of ResNet18 trained with different noise under PGD-10 attack, where GAR stands for a model trained with examples added by normal Gaussian noise, GA stands for training with only global noise, GN stands for training with global noise extended to the feature layers, and GATN stands for training with global noise and attentional noise added. Medical images are from the Dermatology dataset. ϵ denotes the maximum pixels modified by adversarial perturbations and the attack step size is $\alpha = \epsilon/4$, σ represents the initial variance of Gaussian noise.

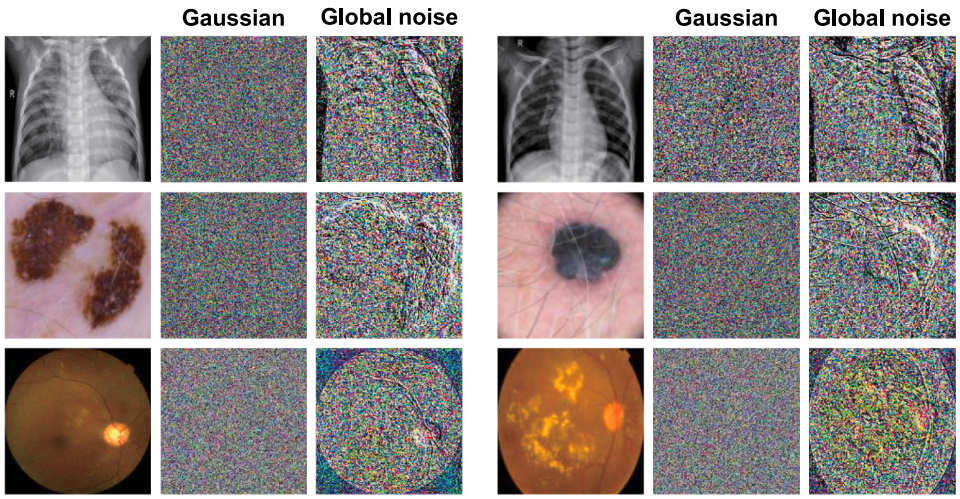


Fig. 4. Qualitative Evaluation of global noise. Our optimized noise adds to the regions where the network robustly learns compared to normal Gaussian noise and leaves other areas untouched.

Table 1
Comparison of the vulnerability of the medical features from different layers on Dermatology dataset. We calculate the change magnitude of the average activation of the mean values of the features from different layers of ResNet-18 under the PGD-10 attack. We use perturbations under the constraint $L_\infty = 8/255$, a bigger change indicates greater vulnerability. The smallest change magnitude are highlighted in bold.

Model	Number of residual block				Penultimate	Logits
	1st	4st	10st	15st		
Standard	0.26%	3.82%	19.77%	56.27%	153.26%	444.73%
GAR	**0.23**%	2.25%	18.55%	**9.16**%	33.10%	60.03%
GA	0.59%	**0.83**%	63.49%	27.73%	32.08%	24.07%
GN	0.55%	3.98%	18.12%	28.83%	**20.34**%	13.45%
GATN	0.78%	14.35%	**6.48**%	15.97%	28.19%	**5.56**%

adversarial examples and calculate the magnitude of their changes. As shown in Table 1, small perturbations in the example layer end up bringing huge changes under the amplification effect of the model.

Second, adversarial examples from different medical image datasets divert the model's attention to different degrees. We find subtle differences in the adversarial examples for the ChestX-ray, Dermatology, and Fundoscopy datasets. For the ChestX-ray dataset, the adversarial perturbation causes the model to lose the lesion features. But for Dermatology and Fundoscopy datasets, the adversarial perturbation can misclassify the lesion features even when the model notices them. This means that attackers focus their attacks according to the intrinsic features of different medical image datasets. We analyzed three datasets and found that they differed in their lesion characteristics. The lesion features of the ChestX-ray dataset are mostly flocculent or shadowed textures with no obvious contours. While the lesions of Fundoscopy are mostly hard exudate as white or yellow dots, which have clear borders as well as the lesions of the Dermatology dataset. Since DNNs tend to use texture rather than shape features [57], the ChestX-ray dataset has more generalization features to exploit by attackers.

The similarity of medical images adversarial examples suggests that we should strengthen the stable intrinsic features and weaken the unstable effects of generalization features, while smoothing the model to reduce the sharp regions. The variability of medical images, motivates us to explore the intrinsic properties of different datasets of medical images.

3.2. Robust with global noise

In this section, we improve the robustness of the model from example perspective. The purpose of global noise imposed on the example layer is to disrupt the perturbation structure while preserving medical image lesion features, thus pushing examples away from sharp regions.

Since Gaussian noise is generally larger in magnitude than the adversarial perturbations, these large random perturbations can swamp the small adversarial perturbations thereby destroying the effect of the adversarial examples on the generalization features [58]. We first add Gaussian noise with fixed parameters to the examples [46]. Fig. 3 shows the robustness of the medical image diagnosis model trained with a different noise. One can observe that fixed random Gaussian noise (GAR) with different parameters brings different effects. The robustness of the model improves as the amplitude of the Gaussian noise increases. However, as the Gaussian noise amplitude increases further, the robust accuracy decreases instead. Because the excessive noise not only drowns out the adversarial perturbation but also drowns out the lesion features.

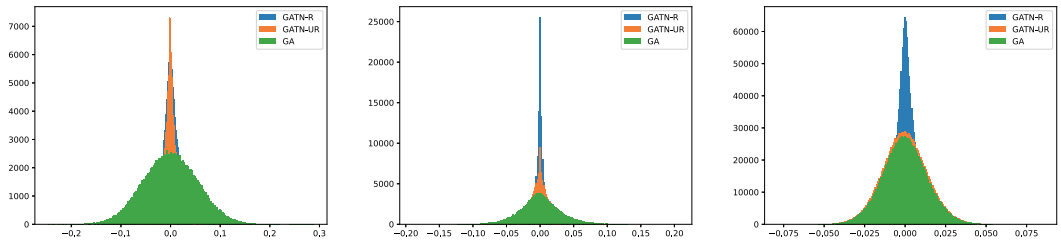
To improve this problem, we add global noise to the example layer and embed it into the model. Considering a set of examples $x \in X$ and their labels $y \in Y$, the medical diagnosis model aims to learn a classification function that maps the examples to the label space $f : X \to Y$. The loss function $L(\cdot)$ is then used to train the model $L(f(x), y; \theta)$, where θ is the weight of the model. A deep neural network is composed of many nonlinear mappings, with each layer corresponding to an output $z_l \in \mathbb{R}^{c\times h\times w}$, where c, h and w represent the number of channels,height and width, respectively.

For a example $x \in \mathbb{R}^{c\times h\times w}$, we add the learnable pixel-level Gaussian noise to the example layer, then:

$$y = f(x + \alpha_x \cdot \eta), \eta \sim N(0, 1) \tag{1}$$

where $\alpha_x \in \mathbb{R}^{c\times h\times w}$ is the trainable noise tensor of the example layer, and η is a Gaussian distribution with a standard deviation of 0 and variance of 1.

Moreover, as shown in Fig. 3, compared with fixed Gaussian noise, global noise (GA) can steadily improve the robustness of the model. Because global noise is independent pixel-level noise, the transformation of each pixel can be carried out independently without affecting the transformation of other pixels, which can make the transformation space more abundant. Since the global noise is trainable and will always decay to the most appropriate parameter regardless of the parameter used as the initial value, global noise is less influenced by the initialization parameters. We choose 0.2 as the noise initial tensor for the balance of clean and robust accuracy. Fig. 4 presents fixed Gaussian noise and global noise. It shows that our proposed global noise can highlight the essential features of the examples. This demonstrates the effect of our defense and explains its effectiveness.



(a) ChestX-ray (b) Dermatology (c) Fundoscopy

Fig. 5. Weight distribution of the 14th convolution layer of a ResNet18 model of three medical datasets for different training schemes. Green histogram represent only global noise (GA) added, blue and orange histogram represent the addition of Global attention lesion-related noise (GATN-R) and Global attention lesion-unrelated noise (GATN-UR), respectively.

We evaluate the robustness of different layers by calculating the change magnitude of the average activation value of the features from each layers in ResNet18. As shown in Table 1, adding only fixed random Gaussian noise (GAR) or global noise (GA) to the example layer mitigate the effect of network amplification. However, the magnitude of change in the top feature layer remains large. Since the features learned in the top layer will directly affect the performance of the model. The tendency for the perturbations to amplify makes it difficult to suppress the adversarial perturbations from the example level alone.

3.3. Robust with attention noise

To address above problem, our approach in this section is to smooth the model. A natural idea is to extend global noise to the feature layers of the model, as shown in Fig. 1 for GN-Net18. However, as illustrated in Fig. 3, mechanically adding global noise to feature layers results in a negligible improvement or even a decrease in accuracy.

This motivates us to choose the right noise instead of the big one. It is more important to improve the noise expression capability than to mechanically increase the noise number. To enhance the noise expression, we apply weights to the feature layer noise. These weights guide the noise added to lesion-related or lesion-unrelated regions according to the characteristics of the medical image datasets. Since the features are extracted by each channel of the model, the noise weights should be consistent with the channel weights. We calculate the channel weights as the weights of the noise.

We define $z_l^{chw} = h_l[c,h,w]$ is the pixel value (h,w) of cth channel feature map, and $z_l^c = z_l[c,:,:]$ is the cth channel feature map tensor. The wights of the channels at layer l can be obtained from the following equation [59]:

$$m_l^c = avg(z_l^c) = \frac{1}{H \cdot W} \sum_{h=1}^{H} \sum_{w=1}^{W} z_l^{chw} \tag{2}$$

$$[\delta_l^1; \ldots; \delta_l^C] = sig(fc([m_l^1; \ldots; m_l^C]; \mathbf{w})) \tag{3}$$

where $c = 1, \ldots, C$ is the number of channels in layer l. First, the average pooling layer is used to compress the channel information to obtain m_l^c, then a linear transformation, a ReLU activation function, and a fully connected layer $fc(\cdot;\mathbf{w})$ with parameter $\mathbf{w}$ are superimposed to fuse all the channel information, and finally a symbolic function $sig(\cdot)$ is used to obtain the scaling factor for each channel.

Our proposed lesion-related attention noise and lesion-unrelated attention noise can be formally described as follow:

$$\widetilde{z}_l^c = f_{GATN-R}(z_l^c) = (z_l^c + \delta_l^c) \cdot \alpha_l^c \cdot \eta, \eta \sim N(0,1) \tag{4}$$

$$\widetilde{z}_l^c = f_{GATN-UR}(z_l^c) = z_l^c + \delta_l^c \cdot z_l^c \cdot \alpha_l^c \cdot \eta, \eta \sim N(0,1) \tag{5}$$

where $\alpha_l^c \in \mathbb{R}^{H \times W}$ is the learnable noise tensor of the feature map of the cth channel in layer l, and η is a Gaussian distribution obeying a standard deviation of 0 and variance of 1, δ_l^c is used to measure the importance of the feature map for that channel and to scale it.

In this paper, we take the noise tensor as a model parameter and optimize it by back-propagation. For the model to heuristically balance clean accuracy and robust accuracy, we use the cross entropy loss as a loss function. Then for the optimization of the attention noise tensor, the gradient calculation can be described as :

$$\frac{\partial L}{\partial \alpha_l} = \sum_c \frac{\partial L}{\partial f_{GATN(z_l^c)}} \cdot \frac{\partial f_{GATN(z_l^c)}}{\partial \alpha_l^c} \tag{6}$$

where $\sum_c$ calculates the noise summation over the c channels, $\partial L / \partial f_{GATN}(x_l^c)$ is the back-propagation of the gradient, and the gradient of $f_{GATNGATN}(\cdot)$ is calculated as :

$$\frac{\partial f_{GATN-R}(z_l^c)}{\partial \alpha_l^c} = z_l^c + \delta_l^c \cdot \eta, \eta \sim N(0,1) \tag{7}$$

$$\frac{\partial f_{GATN-UR}(z_l^c)}{\partial \alpha_l^c} = z_l^c \cdot \delta_l^c \cdot \eta, \eta \sim N(0,1) \tag{8}$$

where z_l^c and δ_l^c are constants in back-propagation, the noise tensor is updated at step t for layer l as :

$$\alpha_l^t = \alpha_l^{t-1} - lr \cdot \frac{\partial L^{t-1}}{\partial \alpha_l} \tag{9}$$

During the optimization process, our proposed attention noise changes its parameters with the loss function. The gradient derivation presents that noise in feature layers changes with the hidden layer features, which makes noise more relevant for medical diagnosis models.

As shown in Table 1, global attention noise effectively suppresses the amplification effect of perturbation in top features layer. We further explore the phenomena in Table 1 in terms of model weights. We investigate how the weight distribution changed before and after applying the feature layer noise. We plotted histograms of weight values for different layers and found that GA-Net18 and GATN-Net18 are similar in the shallower layers, while GATN-Net18 has a smaller magnitude in the deeper layers. Compared with GA model, GATN has more weights converging to zero, and small perturbations added to these features do not cause them to become larger. This indicates feature layer noise locally smoothes the model. Fig. 5 shows the distribution of weight values of ResNet18 in layer 14th for the ChestX-ray, Dermatology, and Fundoscopy datasets.

4. Experiments

4.1. Setup

4.1.1. Datasets

We use three publicly available benchmark datasets for classification tasks. For our model training and attacking experiments, we need two subsets of data for each dataset: (1) subset Train for pre-training the DNN model, and (2) subset Test for evaluating the DNN models and crafting adversarial attacks. The number of classes and images we retrieved from the public datasets can be found in Table 2.

Table 2
Number of classes and images in each subset of three datasets.

Dataset	Classes	Train	Test	Total
ChestX-ray	Normal	1341	253	5863
	Pneumonia	3874	395	
Dermatology	NV	10 578	2162	23 344
	MEL	3768	799	
	BCC	2769	599	
	BKL	2187	482	
Fundoscopy	No DR	1505	295	3662
	DR	1557	305	

ChestX-ray. Kaggle ChestX-ray [60] is mainly used for the classification task of pneumonia. This dataset contains a total of 5863 X-ray images, which is divided into two categories, including pneumonia and normal.

Dermatology. Dermatology dataset is used for the classification task of common pigmented skin diseases. This dataset is from the ISIC-2019 Challenge [61] with a total of 25,331 images and each image is labeled from 'NV' to 'MEL/BCC/BKL/DF/VASC/SCC/UNK'. Referring to [17], to reduce the data imbalance between classes, four classes (NV, MEL, BCC, and BKL), in which the number of images exceeds 2000 images are selected, for a total of 23,344 images.

Fundoscopy. Kaggle Fundoscopy [62] is generally used for the diabetic retinopathy classification task. This dataset contains 3662 images and each image is labeled from 'No DR' to the five levels of 'Mild/Moderate/Severe/Proliferative DR'. Referring to [27,30,63], we transform the dataset into a two-class dataset, in order to find images with a degree above 'Moderate'.

4.1.2. Models and data preprocessing

We evaluate five models on the three datasets: (1) ResNet18 [64], a widely used DNN as our backbone network, (2) GA-Net18, adds only global noise for training, (3) GN-Net18, extends global noise to feature layer, (4) GATN-R, our proposed global lesion-related attention noise for training, and (5) GATN-UR, our proposed global lesion-unrelated attention noise for training. All DNN models in the experiments were trained using Adam, with the initial learning rate set to 5e-4 and the weight decay set to 0.0001. DNN models trained for 100 epochs on the ChestX-ray and Fundoscopy datasets, and for 200 epochs on the Dermatology dataset. All images were centrally cropped to a size of $224 \times 224 \times 3$. Simple data enhancement was used, including rotation, width/height panning, and horizontal flipping. After training was completed, the model was fixed in subsequent adversarial experiments.

4.1.3. Adversarial attack

We used the adversarial examples obtained from attacking ResNet18 to attack each of the other four models and evaluate the robustness by observing the robust accuracy of the model. AA is a combination of APGD, FAB, and SquareAttack, which means it adaptively uses attacks to explore diverse adversarial examples. The attack strength ϵ of BIM, PGD, and AA is $8/255$. For the BIM attack, the steps we set $k = 7$. For the PGD attack, the steps we set $k = 10$ and the α we set to $\epsilon/4$. For C&W attack, the learning rate we set $\alpha = 1{\cdot}10^{-2}$, the number of iteration $k = 1000$, initial constant $c = 1$. For Deepfool, we set the step to 50, and for the initial constant, we set it to 0.02.

4.2. Defend adversarial examples

In this section, we mainly focus on evaluating the robust accuracy of five models against different attacks. In Table 3, we compare five models on ChestX-ray, Dermatology, and Fundoscopy. As shown in Table 3, the original ResNet18 is vulnerable to adversarial attacks on all three datasets (*e.g.*, its accuracy drops to 0% under PGD attack on ChestX-ray, Dermatology, and Fundoscopy). GA-Net18 can significantly improve the robustness of the original model in all attack settings (*e.g.*, under CW attack, the accuracy of GA-Net18 is 89.17% while the original model is 4.17% on Fundoscopy). Another conclusion that can be drawn from Table 3 is that simply adding global noise or inappropriate noise in the feature layers is less likely to improve robust accuracy (*e.g.*, under BIM-5 attack, the accuracy of GA-Net18 on Dermatology is 59.03%, while the accuracy of GN-Net18 and GATN-UR is 55.21% and 57.81%, respectively). Moreover, one can observe that the robust classification results of the three datasets have been significantly improved after using GATN. For instance, on the ChestX-ray dataset, the defense accuracy of GATN-UR is 87.70% when attacked by AA, which is much better than the accuracy of other models. GATN-R and GATN-UR outperform other models, with GATN-UR achieving the best results under all attacks for ChestX-ray and GATN-R achieving the best results for Dermatology and Fundoscopy datasets. Such observation further indicates that GATN-R adds noise in lesion-related areas while GATN-UR adds in lesion-unrelated areas. These results show that our proposed method is effective in converting non-robust models to robust models.

4.3. Defend stronger attacks

In this section, we conduct two stronger attacks: PGD-100 and AA attack. For the PGD-100 attack, the iterative steps are set as 100, which means it uses a smaller step size to explore the adversarial examples. Since too much perturbation can distort medical images, the maximum perturbation we evaluated is $\epsilon = 32/255$. The results are shown in Table 5, even under more powerful attacks or larger perturbations, our proposed method can obtain good robust accuracy. For example, with a perturbation budget of $\epsilon = 32/255$, our best robust accuracy is 67.00% and 57.50% under PGD-100 and AA attack.

4.4. Effect of network capacity

To investigate the relationship between network capacity and robustness improvement by GATN, we evaluated various network architectures in terms of depth. For networks with different depths, experiments on ResNet18/34/50 are conducted under standard training and our proposed method. We report the accuracy of clean and adversarial examples for two feature layers of attention noise. The results in Table 4 show that increasing the capacity of the model indeed improves network robustness against adversarial examples (*e.g.*, the accuracy of ResNet50 on Fundoscopy is 94.33% under AA attack, while ResNet18 and ResNet34 achieve accuracy of 90.83% and 87.83%, respectively). However, larger capacity does not always imply better robustness for medical image models (*e.g.*, the accuracy of ResNet34 on Dermatology is 73.04% under PGD attack while ResNet50 is 71.69%, which is lower than the former). This related to the tendency of medical diagnosis models to overfit. Thereby, the structure for medical diagnosis models still needs to be carefully constructed to prevent overfitting problems caused by parameterization.

4.5. Ablation study

To verify the effectiveness of noise on all layers, we perform an ablation experiment. The robustness accuracy is presented in Table 6. As shown in the first row, the model performs best when we perform correctly defend both the example layer and feature layers.

However, if the noise is partial, it will lead to the degradation of its robustness. We can observe that the global noise in the example layer improves the robustness of the model, comparing the first and second rows. The effect becomes more pronounced as the perturbation budget increases. The same phenomenon can be obtained by comparing the third and fourth rows as well. This is due to the fact that, even after the perturbed structure is eliminated, the examples may still fall into the steep region of the model under the influence of amplification effects.

Table 3
Accuracy (%) of various models against four types of attacks crafted on the ChestX-ray, Dermatology, and Fundoscopy datasets. The maximum L_∞ perturbation is $\epsilon = 8/255$, and the number after the attack method represents the number of iteration steps. The higher accuracy, the better the robustness of the model. The best results are highlighted in bold, and the next best results are underlined.

Datasets	Model	Clean	BIM			PGD			AA	CW	Deepfool
			2	5	7	7	10	100			
ChestX-ray	ResNet18	**93.72**	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.78
	GA-Net18	90.80	82.57	81.42	81.26	83.36	81.79	81.32	83.67	82.80	81.43
	GN-Net18	91.05	81.63	82.57	80.67	83.99	81.00	82.89	84.14	82.41	80.28
	GATN-R	90.42	81.79	82.10	82.89	85.87	82.26	81.63	82.89	83.99	82.64
	GATN-UR	91.68	**87.60**	**87.28**	**87.76**	**88.23**	**86.66**	**87.44**	**87.70**	**86.09**	**88.52**
Dermatology	ResNet18	**85.88**	1.32	1.01	2.28	0.00	0.00	0.00	0.00	0.00	2.95
	GA-Net18	80.71	58.82	59.03	59.29	64.21	59.97	58.39	55.53	68.70	70.29
	GN-Net18	78.97	49.16	55.21	53.43	54.21	61.82	58.20	55.40	66.6	68.71
	GATN-R	80.08	**70.52**	**70.06**	**69.77**	**74.05**	**72.49**	**66.93**	**66.85**	**75.89**	**72.73**
	GATN-UR	80.24	57.81	57.83	57.60	64.21	60.66	57.40	56.46	69.25	64.23
Fundoscopy	ResNet18	**97.67**	11.67	3.33	2.33	4.50	0.00	0.00	0.00	4.17	2.83
	GA-Net18	96.67	83.50	80.00	77.83	90.00	82.33	63.00	61.00	89.17	86.83
	GN-Net18	96.33	84.33	83.50	82.67	88.67	85.67	81.50	76.67	88.33	88.67
	GATN-R	95.67	**92.33**	**91.83**	**90.67**	**92.33**	**90.17**	**86.17**	**87.83**	**91.67**	**91.33**
	GATN-UR	96.83	85.17	82.33	86.00	91.50	84.83	82.67	80.83	90.00	89.50

Table 4
Accuracy (%) on the ChestX-ray, Dermatology, and Fundoscopy, utilizing different robust optimization configurations. For network depth, the classical ResNet18/34/50 with increasing depth is reported.

Dataset	Model	No defense				GATN-R				GARN-UR			
		Clean	BIM	PGD	AA	Clean	BIM	PGD	AA	Clean	BIM	PGD	AA
ChestX-ray	ResNet18	93.72	0.00	0.00	0.00	90.42	82.89	82.26	82.89	91.68	87.76	86.66	87.70
	ResNet34	93.09	0.00	0.00	0.00	90.58	81.32	82.63	81.63	91.84	87.44	87.91	87.13
	ResNet50	93.59	0.00	0.00	0.00	90.80	83.09	83.77	84.91	92.78	86.09	86.19	89.48
Dermatology	ResNet18	85.88	2.28	0.00	0.00	80.08	69.77	72.49	66.85	80.24	57.60	60.66	56.46
	ResNet34	85.60	0.70	0.00	0.00	78.85	68.45	73.04	64.41	80.36	61.31	64.21	57.94
	ResNet50	85.73	0.05	0.00	0.00	79.06	67.38	71.69	61.28	80.06	57.50	57.06	53.54
Fundoscopy	ResNet18	97.67	2.33	0.00	0.00	95.67	90.67	90.17	87.83	96.83	86.00	84.83	80.83
	ResNet34	98.17	0.83	0.00	0.00	95.83	93.00	93.67	90.83	96.50	84.67	85.67	84.67
	ResNet50	98.33	0.00	0.00	0.00	95.50	94.50	95.17	94.33	96.33	84.83	85.50	84.00

Table 5
Evaluating our proposed method with ResNet18 backbone on Fundoscopy, against PGD-100, AA attacks, for different values of attack strength. ϵ denotes the maximum pixels modified by adversarial perturbations and the attack step size of PGD-100 is $\alpha = \epsilon/4$. The best results are highlighted in bold.

Attack	Method	Robustness accuracy (%)					
		$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$
PGD	No defense	5.50	3.17	0.33	0.00	0.00	0.00
	GATN-R	95.17	**95.00**	**92.83**	**87.50**	**72.83**	**67.00**
	GATN-UR	**96.50**	93.17	88.83	77.00	58.00	49.50
AA	No defense	10.00	0.00	0.00	0.00	0.00	0.00
	GATN-R	95.00	94.00	**93.83**	**87.83**	**65.50**	**57.50**
	GATN-UR	**95.67**	**95.17**	93.17	80.83	48.83	40.83

Table 6
Ablation study of global attention noise injection performed on example layer and feature layers. Accuracy (%) on the ChestX-ray, utilizing different robust optimization configurations under PGD-10 attack. ϵ denotes the maximum pixels modified by adversarial perturbations and the attack step size is $\alpha = \epsilon/4$. The best results are highlighted in bold.

Specific Noise			Robustness accuracy (%)				
Global	Lesion-R	Lesion-UR	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$
✓	–	✓	**91.37**	**91.21**	**88.11**	**87.44**	**83.52**
✗	–	✓	90.27	89.01	86.03	81.00	77.71
✓	✓	–	89.32	88.70	84.81	81.63	69.07
✗	✓	–	86.97	86.19	81.16	72.84	50.24
✓	✗	✗	88.27	85.01	82.56	77.53	57.30

Attention noise is also essential. The first and last rows demonstrate that adding only global noise is difficult to suppress the effects of errors caused by larger perturbations. However, adding only model feature layer noise may lead to over-smoothing of the model and poor defense under large perturbation budgets. Therefore both types of noise are important.

In addition, inappropriate feature layer noise can also lead to a decrease in robust accuracy, affecting robust accuracy even more than adding only a single noise (*e.g.*, when the perturbation budget is $\epsilon = 16$, GATN-R (third row) decreases by 14.45%, while lesion-UR (second row) decreases by 5.81%). Moreover, we find that GATN-R is more affected than GATN-UR. Since GATN-R is mainly resistant to perturbations in lesion-related regions, it will be insufficient to resist adversarial examples as the perturbation budget expands both lesion and non-lesion regions. Overall, both components of GATN play their role in improving the robustness of medical image diagnostic models.

4.6. Comparison with other methods

To further validate our method, in this section, we compare GATN with other defense methods, including model-based defense (AT [15], DCN [17], Dual-AT [44], RCNN [49]), example-based defense methods (Random R-P [14], ComDefend [35]). The accuracy of each method can be found in Table 7. From Table 7, we can obtain when the dataset is ChestX-ray, both GATN-R and GATN-UR have high defense accuracy. For example, GATN-R and GATN-UR have 83.99% and 86.09% after being attacked by C&W, while DCN, a defense method for medical images, has an accuracy of 80.54%. For Dermatology, GATN-UR achieves better accuracy under many attacks and GATN-R achieves the highest robust accuracy under all attack methods. GATN maintains the best defense accuracy for Fundoscopy under many attacks (*e.g.*, PGD, CW, and Deepfool). Our method outperforms example-based and model-based methods for natural images, as well as defense methods for medical images. Furthermore, we employ clean examples for training, resulting in higher clean accuracy.

Table 7
Accuracy (%) of GATN compared with other defense methods under different adversarial attacks on the ChestX-ray, Dermatology, and Fundoscopy. The maximum L_∞ perturbation is $\epsilon = 8/255$, and the number after the attack method represents the number of iteration steps. The higher accuracy, the better the robustness of the model. The best results are highlighted in bold, and the next best results are underlined.

Dataset	Model	Method	Clean	BIM		PGD		AA		CW	Deepfool
				$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 4$	$\epsilon = 8$		
ChestX-ray	ResNet18	No defense	**93.72**	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		AT [15]	86.03	77.08	68.86	76.92	68.76	70.53	62.47	73.83	72.42
		Dual-AT [44]	89.86	81.42	75.53	80.53	74.93	75.81	70.59	76.91	78.84
		Random R-P [14]	62.48	59.81	55.89	59.65	50.45	50.32	44.92	61.94	60.98
		ComDefend [35]	81.68	65.93	63.27	64.99	61.22	63.59	60.89	72.37	74.32
		DCN [17]	89.95	84.93	82.26	85.14	81.69	84.40	82.10	80.54	83.33
		RCNN [49]	88.60	80.85	73.24	78.11	72.09	75.63	68.06	81.37	82.70
		GATN-R	90.42	86.81	81.49	86.19	82.26	84.14	82.89	83.99	82.64
		GATN-UR	91.68	**89.64**	**87.76**	**89.17**	**86.66**	**88.54**	**87.70**	**86.09**	**88.52**
Dermatology	ResNet18	No defense	**85.88**	2.21	2.28	0.00	0.00	0.00	0.00	0.00	2.95
		AT [15]	76.95	63.51	60.73	63.48	61.76	52.94	45.77	65.79	60.68
		Dual-AT [44]	79.64	65.73	59.78	63.59	58.45	59.32	53.65	67.01	65.46
		Random R-P [14]	67.31	48.22	43.46	45.93	41.49	40.38	35.42	46.41	48.82
		ComDefend [35]	72.78	62.66	54.59	60.25	52.75	58.60	55.72	66.83	64.48
		DCN [17]	79.80	60.42	51.07	62.96	55.48	48.42	40.95	53.59	58.35
		RCNN [49]	78.23	62.71	56.45	60.48	54.08	55.28	52.16	65.36	64.54
		GATN-R	80.08	**74.41**	**69.77**	**74.62**	**72.49**	**70.24**	**66.85**	**75.89**	**72.73**
		GATN-UR	80.24	66.59	57.60	67.00	60.66	66.38	56.46	69.25	64.23
Fundoscopy	ResNet18	No defense	**97.67**	17.00	2.33	0.50	0.00	0.00	0.00	4.17	2.83
		AT [15]	92.67	87.17	83.50	85.67	82.33	80.83	70.00	86.33	85.17
		Dual-AT [44]	94.17	88.33	87.67	84.85	83.50	82.17	78.00	85.33	83.67
		Random R-P [14]	85.33	69.50	64.17	68.33	62.83	61.67	56.00	73.50	70.67
		ComDefend [35]	91.85	77.83	72.00	71.17	67.17	70.50	71.17	82.17	83.33
		DCN [17]	97.17	92.67	**91.67**	90.00	85.33	90.33	86.50	89.33	88.33
		RCNN [49]	93.50	90.17	87.50	89.33	86.00	81.67	75.33	84.50	86.67
		GATN-R	95.67	93.83	90.67	**93.67**	**90.17**	93.83	**87.83**	**91.67**	**91.33**
		GATN-UR	96.83	**94.00**	86.00	91.33	84.83	**94.17**	80.83	90.00	87.50

5. Discussion

In this section, we aim to explore the effect of GATN on the model in terms of loss landscape and human vision alignment.

The loss landscape provides strong evidence for the smoothness of the model. By exploring the effect of local variations around individual input example on the loss function, we can observe how our method brings changes to the model. As in previous work in medical images [30], we construct two adversarial directions g and $g^{\perp}$, where g and $g^{\perp}$ are gradients extracted from medical image diagnosis models and a set of separately trained agent models. Then we generate adversarial examples following $x_{adv} = x + \epsilon_1 \cdot g + \epsilon_2 \cdot g^{\perp}$ and gradually increase ϵ_1 and ϵ_2 from 0 to 8/255. From Fig. 7, we observe that a slight perturbation in a medical image can lead to a sharp increase in the loss landscape which leads to the vulnerability of the medical diagnosis model. By adding GATN, the loss landscape of the adversarially trained model is much flatter and the abrupt changes disappear. We can consider that the model is relatively insensitive to perturbations at this point.

Finally, we explain the robustness of the model in terms of alignment to human visual perception, such that a stronger model gains a more semantically meaningful gradient. As shown in Fig. 6, gradients for the GATN-trained model align well with the perceptually relevant feature of the input images, containing more contours associated with the lesion. In contrast, the semantic information extracted by the standard-trained model does not correlate well with the lesion information and is easily disturbed by perturbations. Moreover, The difference between GATN-R and GATN-UR is not significant for the ChestX-ray. However, for Dermatology and Fundoscopy datasets, GATN-R and GATN-UR showed greater differences. Although both modalities enhanced the contour information of the lesion area, the location of the enhancement differed. The lesion-related noise (GATN-R) enhances the gradient information inside the lesion area, whereas the lesion-unrelated noise (GATN-UR) mainly enhances the gradient information outside the lesion area, and thus explains the lack of robustness of GATN-UR in Dermatology and Fundoscopy datasets.

6. Conclusion

We propose a simple but efficient defense method, GATN, aiming to protect medical imaging systems against various types of attack methods. Our defense is based on the key observation that Attackers generate adversarial perturbations based on the characteristics of different medical image datasets. These perturbations are applied to medical images in order to shift the attention of the medical image classification model. Based on these experimental results, the main conclusion can be summarized as follows: First, We proposed GATN which relies on two components for defense: global noise in the example layer and attention noise in the feature layers. Global noise can destroy the adversarial perturbation structure and keep the examples away from sharp loss regions that are vulnerable to attack. The attention noise can locally smooth the model. To be specific, for the Chest X-ray dataset with unclear lesion boundaries we propose lesion-unrelated attention noise (GATN-UR), and for the Dermatology and Fundoscopy datasets with clearer lesion boundaries we propose lesion-related attention noise (GATN-R). Second, we perform our experiment by using ResNet18. The results confirm that Both components combined build a lightweight but very effective defense against various attacks, including gradient-based BIM, PGD, and AA, as well as optimization-based C&W and Deepfool.

However, GATN is limited by the structure of the model. We hope that the same effect of smooth models under random noise can be achieved in future work with other attention noise that do not rely on model structure. In addition to this, we will explore whether this approach can be combined with existing defense methods to obtain better performance. we will also explore whether the method can be applied to other tasks in the medical image domain, such as segmentation tasks.

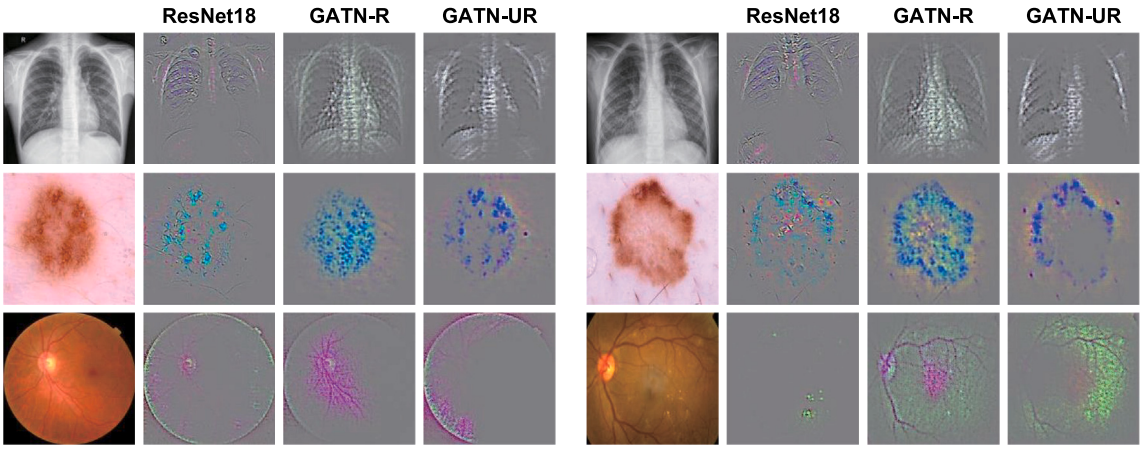


Fig. 6. Qualitative Evaluation of attention noise. Our optimized noise adds to the lesion-related regions and lesion-unrelated regions according to the characteristic of different medical image datasets.

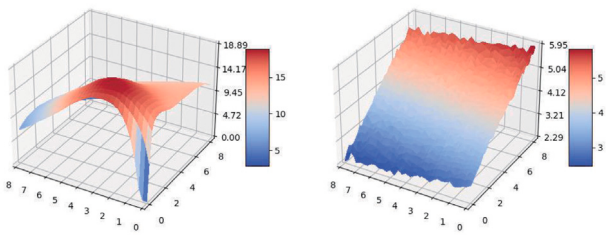


Fig. 7. The loss landscape of ResNet18(a) and GTAN-R (b) in Dermatology dataset. The $x, y - axis$ of the loss landscape plots are ϵ_1 and ϵ_2, which are the sizes perturbations. The $z - axis$ of the loss landscape is the classification loss.

Declaration of competing interest

None Declared.

Acknowledgments

This work was supported by the Major Research Plan of the National Natural Science Foundation of China (92167203), Key Program of Zhejiang Provincial Natural Science Foundation of China (LZ22F020007), the Foundation of Zhejiang Key Laboratory of Multidimensional Perception Technology Application and Cybersecurity, China (HIK2022001), the Major Key Project of Peng Cheng Laboratory (2022A03), and the Science and Technology Innovation Foundation for Graduate Students of Zhejiang University of Science and Technology, China (F464108M05).

References

[1] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, et al., Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning, 2017, arXiv preprint arXiv:1711.05225.

[2] C. Barata, M.E. Celebi, J.S. Marques, Explainable skin lesion diagnosis using taxonomies, Pattern Recognit. 110 (2021) 107413.

[3] N. Tsiknakis, D. Theodoropoulos, G. Manikis, E. Ktistakis, O. Boutsora, A. Berto, F. Scarpa, A. Scarpa, D.I. Fotiadis, K. Marias, Deep learning for diabetic retinopathy detection and classification based on fundus images: A review, Comput. Biol. Med. 135 (2021) 104599.

[4] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, R. Fergus, Intriguing properties of neural networks, in: 2nd International Conference on Learning Representations, ICLR 2014, 2014.

[5] A. Kurakin, I. Goodfellow, S. Bengio, Adversarial machine learning at scale, 2016, arXiv preprint arXiv:1611.01236.

[6] M. Paschali, S. Conjeti, F. Navarro, N. Navab, Generalizability vs. robustness: investigating medical imaging networks using adversarial examples, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2018, pp. 493–501.

[7] M. Xu, T. Zhang, Z. Li, M. Liu, D. Zhang, Towards evaluating the robustness of deep diagnostic models by adversarial attack, Med. Image Anal. 69 (2021) 101977.

[8] G. Bortsova, C. González-Gonzalo, S.C. Wetstein, F. Dubost, I. Katramados, L. Hogeweg, B. Liefers, B. van Ginneken, J.P. Pluim, M. Veta, et al., Adversarial attack vulnerability of medical image analysis systems: Unexplored factors, Med. Image Anal. 73 (2021) 102141.

[9] S.G. Finlayson, J.D. Bowers, J. Ito, J.L. Zittrain, A.L. Beam, I.S. Kohane, Adversarial attacks on medical machine learning, Science 363 (6433) (2019) 1287–1289.

[10] D. Kalimeris, G. Kaplun, P. Nakkiran, B. Edelman, T. Yang, B. Barak, H. Zhang, Sgd on neural networks learns functions of increasing complexity, Adv. Neural Inf. Process. Syst. 32 (2019).

[11] S.-M. Moosavi-Dezfooli, A. Fawzi, J. Uesato, P. Frossard, Robustness via curvature regularization, and vice versa, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 9078–9086.

[12] A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, A. Madry, Adversarial examples are not bugs, they are features, Adv. Neural Inf. Process. Syst. 32 (2019).

[13] E. Raff, J. Sylvester, S. Forsyth, M. McLean, Barrage of random transforms for adversarially robust defense, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 6521–6530.

[14] C. Xie, J. Wang, Z. Zhang, Z. Ren, A. Yuille, Mitigating adversarial effects through randomization, in: International Conference on Learning Representations, 2018.

[15] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, A. Vladu, Towards deep learning models resistant to adversarial attacks, in: International Conference on Learning Representations, 2018.

[16] A. Shafahi, A. Ghiasi, F. Huang, T. Goldstein, Label smoothing and logit squeezing: a replacement for adversarial training? 2019, arXiv preprint arXiv:1910.11585.

[17] F.-F. Xue, J. Peng, R. Wang, Q. Zhang, W. Zheng, Improving robustness of medical image diagnosis with denoising convolutional neural networks, in: MICCAI, 2019.

[18] M. Xu, T. Zhang, D. Zhang, Medrdf: a robust and retrain-less diagnostic framework for medical pretrained models against adversarial attack, IEEE Trans. Med. Imaging 41 (8) (2022) 2130–2143.

[19] I. Wasserman, Adversarially robust medical classification via attentive convolutional neural networks, 2022, ArXiv abs/2210.14405.

[20] O.N. Manzari, H. Ahmadabadi, H. Kashiani, S.B. Shokouhi, A. Ayatollahi, MedViT: A robust vision transformer for generalized medical image classification, Comput. Biol. Med. (2023) 106791.

[21] I.J. Goodfellow, J. Shlens, C. Szegedy, Explaining and harnessing adversarial examples, 2014, CoRR abs/1412.6572.

[22] A. Kurakin, I.J. Goodfellow, S. Bengio, Adversarial examples in the physical world, in: Artificial Intelligence Safety and Security, Chapman and Hall/CRC, 2018, pp. 99–112.

[23] F. Croce, M. Hein, Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks, in: International Conference on Machine Learning, PMLR, 2020, pp. 2206–2216.

[24] N. Carlini, D.A. Wagner, Towards evaluating the robustness of neural networks, in: 2017 IEEE Symposium on Security and Privacy (SP), 2016, pp. 39–57.

[25] S.-M. Moosavi-Dezfooli, A. Fawzi, P. Frossard, Deepfool: a simple and accurate method to fool deep neural networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2574–2582.

[26] G. Qi, L. Gong, Y. Song, K. Ma, Y. Zheng, Stabilized medical image attacks, in: International Conference on Learning Representations, 2021, URL: https://openreview.net/forum?id=QfTXQiGYudJ.

[27] Q. Yao, Z. He, Y. Lin, K. Ma, Y. Zheng, S.K. Zhou, A hierarchical feature constraint to camouflage medical adversarial attacks, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part III 24, Springer, 2021, pp. 36–47.

[28] Q. Zhou, M. Zuley, Y. Guo, L. Yang, B. Nair, A. Vargo, S. Ghannam, D. Arefan, S. Wu, A machine and human reader study on AI diagnosis model safety under attacks of adversarial images, Nature Commun. 12 (1) (2021) 7281.

[29] Z. Wang, X. Shu, Y. Wang, Y. Feng, L. Zhang, Z. Yi, A feature space-restricted attention attack on medical deep learning systems, IEEE Trans. Cybern. (2022).

[30] X. Ma, Y. Niu, L. Gu, Y. Wang, Y. Zhao, J. Bailey, F. Lu, Understanding adversarial attacks on deep learning based medical image analysis systems, Pattern Recognit. 110 (2021) 107332.

[31] H. Zheng, J. Chen, H. Du, W. Zhu, S. Ji, X. Zhang, Grip-gan: An attack-free defense through general robust inverse perturbation, IEEE Trans. Dependable Secure Comput. 19 (6) (2021) 4204–4224.

[32] Y. Gong, Y. Yao, Y. Li, Y. Zhang, X. Liu, X. Lin, S. Liu, Reverse engineering of imperceptible adversarial image perturbations, 2022, arXiv preprint arXiv:2203.14145.

[33] Y. Wang, Y. Li, Z. Shen, Fight fire with fire: Reversing skin adversarial examples by multiscale diffusive and denoising aggregation mechanism, 2022, arXiv preprint arXiv:2208.10373.

[34] F. Liao, M. Liang, Y. Dong, T. Pang, X. Hu, J. Zhu, Defense against adversarial attacks using high-level representation guided denoiser, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 1778–1787.

[35] X. Jia, X. Wei, X. Cao, H. Foroosh, Comdefend: An efficient image compression model to defend adversarial examples, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 6084–6092.

[36] W. Nie, B. Guo, Y. Huang, C. Xiao, A. Vahdat, A. Anandkumar, Diffusion models for adversarial purification, in: International Conference on Machine Learning, PMLR, 2022, pp. 16805–16827.

[37] K. Kansal, P.S. Krishna, P.B. Jain, R. Surya, P. Honnavalli, S. Eswaran, Defending against adversarial attacks on Covid-19 classifier: A denoiser-based approach, Heliyon 8 (10) (2022) e11209.

[38] L.D. Le, H. Fu, X. Xu, Y. Liu, Y. Xu, J. Du, J.T. Zhou, R. Goh, An efficient defending mechanism against image attacking on medical image segmentation models, in: Resource-Efficient Medical Image Analysis: First MICCAI Workshop, REMIA 2022, Singapore, September 22, 2022, Proceedings, Springer, 2022, pp. 65–74.

[39] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, Springer, 2015, pp. 234–241.

[40] H. Wang, S. Wang, Z. Qin, Y. Zhang, R. Li, Y. Xia, Triple attention learning for classification of 14 thoracic diseases using chest radiography, Med. Image Anal. 67 (2021) 101846.

[41] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, M. Jordan, Theoretically principled trade-off between robustness and accuracy, in: International Conference on Machine Learning, PMLR, 2019, pp. 7472–7482.

[42] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, Q. Gu, Improving adversarial robustness requires revisiting misclassified examples, in: International Conference on Learning Representations, 2020.

[43] S. Liu, A.A.A. Setio, F.C. Ghesu, E. Gibson, S. Grbic, B. Georgescu, D. Comaniciu, No surprises: Training robust lung nodule detection for low-dose CT scans by augmenting with adversarial attacks, IEEE Trans. Med. Imaging 40 (1) (2020) 335–345.

[44] T. Han, S. Nebelung, F. Pedersoli, M. Zimmermann, M. Schulze-Hagen, M. Ho, C. Haarburger, F. Kiessling, C. Kuhl, V. Schulz, et al., Advancing diagnostic performance and clinical usability of neural networks via adversarial training and dual batch normalization, Nature Commun. 12 (1) (2021) 4315.

[45] S. Sun, M. Xian, A. Vakanski, H. Ghanem, MIRST-DM: Multi-instance RST with drop-max layer for robust classification of breast cancer, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September 18–22, 2022, Proceedings, Part IV, Springer, 2022, pp. 401–410.

[46] S. Zheng, Y. Song, T. Leung, I.J. Goodfellow, Improving the robustness of deep neural networks via stability training, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 4480–4488.

[47] A.S. Rakin, Z. He, D. Fan, Parametric noise injection: Trainable randomness to improve deep neural network robustness against adversarial attack, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 588–597.

[48] A. Jeddi, M.J. Shafiee, M. Karg, C. Scharfenberger, A. Wong, Learn2Perturb: An end-to-end feature perturbation learning to improve adversarial robustness, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 1238–1247.

[49] X. Shi, Y. Peng, Q. Chen, T. Keenan, A.T. Thavikulwat, S. Lee, Y. Tang, E.Y. Chew, R.M. Summers, Z. Lu, Robust convolutional neural networks against adversarial attacks on medical images, Pattern Recognit. 132 (2022) 108923.

[50] Z. Wu, G. Li, S. Shen, X. Lian, E. Chen, G. Xu, Constructing dummy query sequences to protect location privacy and query privacy in location-based services, World Wide Web 24 (2020) 25–49.

[51] Z. Wu, S. Shen, X. Lian, X. Su, E. Chen, A dummy-based user privacy protection approach for text information retrieval, Knowl.-Based Syst. 195 (2020) 105679.

[52] S.-W. Chen, D.-L. Chiang, C.-H. Liu, T.-S. Chen, F. Lai, H. Wang, W. Wei, Confidentiality protection of digital health records in cloud computing, J. Med. Syst. 40 (2016) 1–12.

[53] Z. Wu, S. Xuan, J. Xie, C. Lin, C. Lu, How to ensure the confidentiality of electronic medical records on the cloud: A technical perspective, Comput. Biol. Med. 147 (2022) 105726.

[54] X. Zhang, J. Wang, T. Wang, R. Jiang, J. Xu, L. Zhao, Robust feature learning for adversarial defense via hierarchical feature alignment, Inform. Sci. 560 (2021) 256–270.

[55] C. Xie, Y. Wu, L.v.d. Maaten, A.L. Yuille, K. He, Feature denoising for improving adversarial robustness, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 501–509.

[56] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 618–626.

[57] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F.A. Wichmann, W. Brendel, ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness, in: International Conference on Learning Representations, 2018.

[58] J.M. Cohen, E. Rosenfeld, J.Z. Kolter, Certified adversarial robustness via randomized smoothing, in: International Conference on Machine Learning, 2019.

[59] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.

[60] Kaggle: Chest X-Ray images (pneumonia), 2019, https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

[61] ISIC: The international skin imaging collaboration, 2019, https://www.isic-archive.com.

[62] Kaggle: Aptos 2019 blindness detection, 2019, https://www.kaggle.com/datasets/mariaherrerot/aptos2019.

[63] S.G. Finlayson, H.W. Chung, I.S. Kohane, A.L. Beam, Adversarial attacks against medical deep learning systems, 2018, arXiv preprint arXiv:1804.05296.

[64] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.